

Sentiment Analysis — Summary

Amazon Food Reviews Dataset | Project Report

1. Overview

This project performs sentiment analysis on approximately 5,000 Amazon food product reviews using the TextBlob natural language processing library. Each review was classified as Positive, Negative, or Neutral based on its polarity score. The dataset includes both the full review text and a numerical star rating (Score) from 1 to 5.

2. Key Findings

Sentiment	Approx. Count	Percentage
Positive	~3,500+	~70%+
Negative	~700+	~15%
Neutral	~600+	~12%

3. What Customers Complain About

Negative reviews in this dataset frequently mention **taste** and **product quality** as the core concerns. Customers expressed dissatisfaction when the product did not meet their taste expectations or when the overall product quality was below what was described. These two themes — taste and product — appeared most frequently across all negative reviews, making them the primary complaint areas.

4. Surprising Finding

The most surprising finding was that **88.32% of reviews were Positive**, which is an unusually high satisfaction rate for an e-commerce food dataset. Additionally, some reviews with low star ratings (Score 1–2) were still classified as **Neutral** by TextBlob, meaning those customers expressed their disappointment in a calm and factual tone rather than negative language. This shows that star ratings and sentiment scores do not always tell the same story — a customer can be unhappy but still write in a neutral way.

5. Business Recommendation

Based on the analysis, since 88.32% of reviews are positive, the business is performing well overall. However, to reduce the 10.08% negative reviews, the business should focus on improving product taste and overall product quality, as these are the most frequently mentioned concerns in negative reviews. Ensuring consistent taste and better quality control can help convert dissatisfied customers into positive ones.